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1. Problem Statement

Analyze the large Chicago traffic crash dataset using Hadoop MapReduce and Apache Pig based on different data fields to identify useful crash patterns and insights.

2. Dataset Description

Dataset : <https://data.cityofchicago.org/Transportation/Traffic-Crashes-Crashes/85ca-t3if>

Dataset Name : Traffic Crashes – Crashes

Source : City of Chicago Data Portal

Original Records: 1,092,721 crash records

Original Attributes: 49 columns

3.Dataset Preparation

The original Chicago Traffic Crashes dataset contained 1,092,721 crash records with 49 attributes. For this project, 24 relevant attributes were selected based on the analysis requirements. A Python script was used to extract these selected fields and create a reduced dataset. After column selection, another Python script was used to clean and convert the dataset into a Hadoop-compatible CSV format. The cleaning process included converting the | separator into commas, replacing commas inside values with semicolons, removing line breaks, and trimming extra spaces.

The final prepared dataset containing 24 attributes was saved as ChicagoTrafficCrashes.csv and used for Hadoop MapReduce and Apache Pig analysis.

```
import csv

input_file = "C:/BDAL/BDAPProject/Chicago_Traffic_Crashes.csv"
output_file = "C:/BDAL/BDAPProject/Chicago_Crashes_Hadoop.csv"

selected_columns = [
    "CRASH_RECORD_ID", "CRASH_DATE", "POSTED_SPEED_LIMIT", "TRAFFIC_CONTROL_DEVICE",
    "DEVICE_CONDITION", "WEATHER_CONDITION", "LIGHTING_CONDITION", "FIRST_CRASH_TYPE",
    "TRAFFICWAY_TYPE", "ROADWAY_SURFACE_COND", "ROAD_DEFECT", "CRASH_TYPE", "DAMAGE",
    "PRIM_CONTRIBUTORY_CAUSE", "STREET_DIRECTION", "STREET_NAME", "NUM_UNITS",
    "MOST_SEVERE_INJURY", "INJURIES_TOTAL", "INJURIES_FATAL", "INJURIES_INCAPACITATING",
    "CRASH_HOUR", "CRASH_DAY_OF_WEEK", "CRASH_MONTH"
]

rows = 0

with open(input_file, "r", encoding="utf-8-sig", newline="") as fin, \
    open(output_file, "w", encoding="utf-8", newline="") as fout:

    reader = csv.DictReader(fin)

    writer = csv.DictWriter(
        fout,
        fieldnames=selected_columns,
        delimiter="|"
    )

    writer.writeheader()

    for row in reader:
        selected_row = {
            column: row[column]
            for column in selected_columns
        }

        writer.writerow(selected_row)
        rows += 1

print("Done!")
print("Selected columns:", len(selected_columns))
print("Total data rows:", rows)
print("Created:", output_file)
```

Figure: Python Script for Selecting 24 Required Columns

```

import csv

input_file = "C:/BDAL/BDAPProject/Chicago_Crashes_Hadoop.csv"
output_file = "C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv"
rows = 0
with open(input_file, "r", encoding="utf-8", newline="") as fin, \
    open(output_file, "w", encoding="utf-8", newline="") as fout:

    reader = csv.reader(fin, delimiter="|")
    writer = csv.writer(
        fout,
        delimiter=";",
        quoting=csv.QUOTE_MINIMAL
    )

    for row in reader:

        cleaned = []

        for value in row:
            value = value.replace(",", ";")
            value = value.replace("'", '"')
            value = value.replace("\n", " ")
            value = value.replace("\r", " ")

            cleaned.append(value.strip())

        writer.writerow(cleaned)
        rows += 1

print("Done!")
print("Total rows including header:", rows)
print("Created:", output_file)

```

Figure: Python Script for Cleaning and Converting the Selected Dataset

Field	Description
CRASH_RECORD_ID	A unique ID for each traffic crash
CRASH_DATE	Date and time when the crash occurred
POSTED_SPEED_LIMIT	Speed limit of the road where the crash occurred
TRAFFIC_CONTROL_DEVICE	Type of traffic control device, such as traffic signal or stop sign
DEVICE_CONDITION	Condition of the traffic control device
WEATHER_CONDITION	Weather condition at the time of the crash
LIGHTING_CONDITION	Lighting condition at the crash location

FIRST_CRASH_TYPE	Type of the first collision
TRAFFICWAY_TYPE	Type of road or trafficway
ROADWAY_SURFACE_COND	Condition of the road surface
ROAD_DEFECT	Road defect present at the crash location
CRASH_TYPE	General classification of the crash
DAMAGE	Estimated level of property damage
PRIM_CONTRIBUTORY_CAUSE	Main contributing cause of the crash
STREET_DIRECTION	Direction of the street
STREET_NAME	Name of the street where the crash occurred
NUM_UNITS	Number of vehicles or units involved in the crash
MOST_SEVERE_INJURY	Most serious injury reported in the crash
INJURIES_TOTAL	Total number of injured people
INJURIES_FATAL	Number of fatal injuries
INJURIES_INCAPACITATING	Number of serious or incapacitating injuries
CRASH_HOUR	Hour of the day when the crash occurred
CRASH_DAY_OF_WEEK	Day of the week when the crash occurred
CRASH_MONTH	Month when the crash occurred

4. Hadoop Distributed File System Setup

Before performing the analysis, Hadoop services were started and checked using the command line.

The Hadoop Distributed File System was used to store the dataset. The dataset was placed inside the /chicago/input directory so that it could be processed by MapReduce jobs.

The following commands were used during setup:

```
start-all.cmd
jps
hdfs dfs -ls /
```

```

hdfs dfs -mkdir /Chicago
hdfs dfs -mkdir /chicago/input
hdfs dfs -ls /Chicago
hdfs dfs -put "C:\BDAL\BDAProject\ChicagoTrafficCrashes.csv" /chicago/input/
hdfs dfs -ls /chicago/input
hdfs dfs -mkdir /chicago/output
hdfs dfs -ls /Chicago

```

The jps command was used to check whether important Hadoop services such as NameNode, DataNode, ResourceManager and NodeManager were running correctly.

Hadoop Overview Datanodes Datanode Volume Failures Snapshot Startup Progress Utilities ▾

Browse Directory

/chicago

Show entries Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	drwxr-xr-x	Biswas	supergroup	0 B	Sep 10 13:05	0	0 B	input	
☐	drwxr-xr-x	Biswas	supergroup	0 B	Sep 10 14:30	0	0 B	output	

Showing 1 to 2 of 2 entries

Hadoop, 2026.

Hadoop Overview Datanodes Datanode Volume Failures Snapshot Startup Progress Utilities ▾

Browse Directory

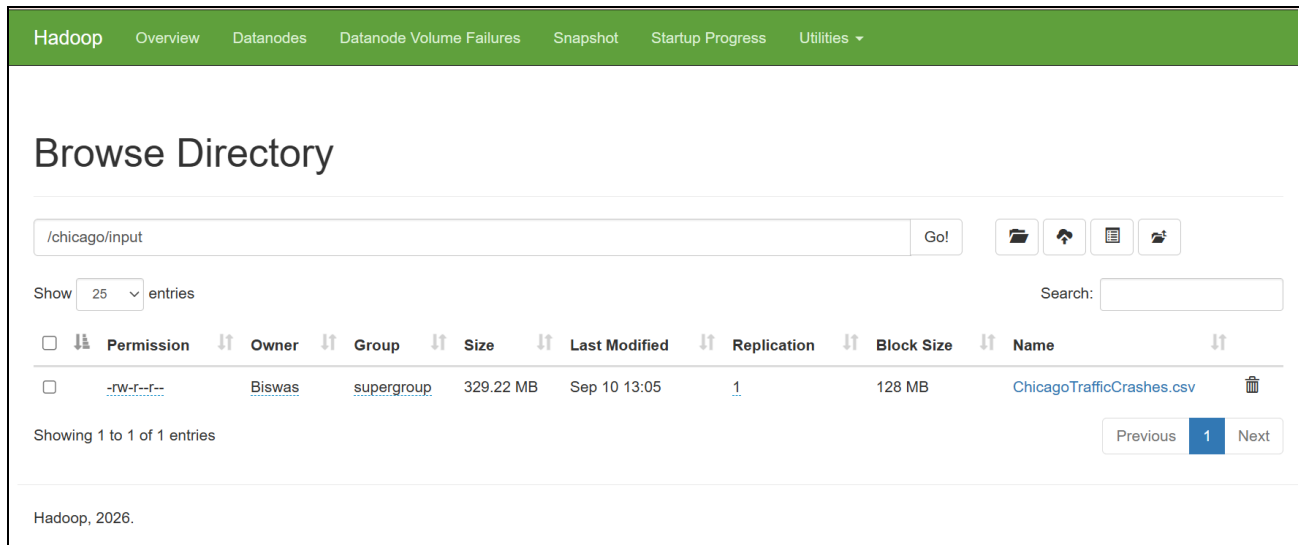
/chicago

Show entries Search:

☐	Permission	Owner	Group	Size	Last Modified	Replication	Block Size	Name	
☐	drwxr-xr-x	Biswas	supergroup	0 B	Sep 10 13:05	0	0 B	input	
☐	drwxr-xr-x	Biswas	supergroup	0 B	Sep 10 14:30	0	0 B	output	

Showing 1 to 2 of 2 entries

Hadoop, 2026.



5. Summary of Data Analysis

In this project, six MapReduce analyses and eight Pig analyses were performed.

MapReduce Analysis

1. Weather-Wise Crash Count
2. Road Surface-Wise Total Injuries
3. Crash Type-Wise Average Speed
4. Road Surface-Wise Injury Rate
5. High-Risk Condition Score
6. Traffic Control-Wise Severe Crash Rate

Pig Analysis

1. Top 10 Accident-Prone Streets
2. Top 10 Primary Causes of Crashes
3. Top 10 Dangerous Hours by Total Injuries
4. Monthly Crash Trend
5. Day-of-Week Crash Analysis
6. Top Crash Types
7. Fatality Ranking by Crash Type and Road Condition
8. Most Severe Injury Distribution

These analyses cover different types of big-data operations such as filtering, grouping, counting, summation, averaging, percentage calculation, sorting, ranking and multi-field analysis.

6. MapReduce Analysis

6.1 Weather-Wise Crash Count

Weather-Wise Crash Count: This analysis calculates the total number of traffic crashes recorded under each weather condition.

This analysis comprises of:

1. **Mapper Class:** Reads each crash record, extracts the weather condition and emits the weather condition with a value of 1.
2. **Reducer Class:** Adds all the values belonging to the same weather condition and returns the total number of crashes.
3. **Driver Class:** Configures the Mapper and Reducer classes and defines the input and output paths for the Hadoop job.

Mapper Class

```
public static class CrashMapper
    extends Mapper<Object, Text, Text, IntWritable> {

    private final static IntWritable one = new IntWritable(1);
    private Text weather = new Text();

    public void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();
        if (line.startsWith("CRASH_RECORD_ID")) {
            return;
        }
        String[] fields = line.split(",", -1);

        if (fields.length != 24) {
            return;
        }
        String weatherCondition = fields[5].trim();

        if (!weatherCondition.isEmpty()) {
            weather.set(weatherCondition);
            context.write(weather, one);
        }
    }
}
```

Reducer Class

```
public static class CrashReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    private IntWritable result = new IntWritable();

    public void reduce(Text key,
        Iterable<IntWritable> values,
        Context context)
        throws IOException, InterruptedException {

        int sum = 0;

        for (IntWritable value : values) {
            sum += value.get();
        }

        result.set(sum);
        context.write(key, result);
    }
}
```

Driver Class

```
public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    Job job = Job.getInstance(
        conf,
        "Weather Wise Crash Frequency Analysis"
    );

    job.setJarByClass(WeatherWiseCrash.class);
    job.setMapperClass(CrashMapper.class);
    job.setReducerClass(CrashReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(
        job,
        new Path(args[0])
    );

    FileOutputFormat.setOutputPath(
        job,
        new Path(args[1])
    );

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
```

Execution

```
Administrator: Command Prompt

C:\BDAL\BDAPProject>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*;%HADOOP_HOME%\share\hadoop\mapreduce\*" -d . WeatherWiseCrash.java

C:\BDAL\BDAPProject>jar -cvf WeatherWiseCrash.jar WeatherWiseCrash.class WeatherWiseCrash$CrashMapper.class WeatherWiseCrash$CrashReducer.class
added manifest
adding: WeatherWiseCrash.class(in = 1497) (out= 811)(deflated 45%)
adding: WeatherWiseCrash$CrashMapper.class(in = 1944) (out= 852)(deflated 56%)
adding: WeatherWiseCrash$CrashReducer.class(in = 1758) (out= 744)(deflated 57%)

C:\BDAL\BDAPProject>hadoop jar WeatherWiseCrash.jar WeatherWiseCrash /chicago/input/ChicagoTrafficCrashes.csv /chicago/output/weather_wise_crash
2026-09-10 13:32:26,135 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 13:32:27,353 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with
ToolRunner to remedy this.
2026-09-10 13:32:27,361 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Biswas/.staging/job_1789023718167_0001
2026-09-10 13:32:27,879 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 13:32:28,007 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 13:32:28,281 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789023718167_0001
2026-09-10 13:32:28,281 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 13:32:28,496 INFO conf.Configuration: resource-types.xml not found
2026-09-10 13:32:28,497 INFO resource.ResourceUtil: Unable to find 'resource-types.xml'.
2026-09-10 13:32:29,009 INFO impl.YarnClientImpl: Submitted application application_1789023718167_0001
2026-09-10 13:32:29,078 INFO mapreduce.Job: The url to track the job: http://DESKTOP-JCQRA5T:8088/proxy/application_1789023718167_0001/
2026-09-10 13:32:29,081 INFO mapreduce.Job: Running job: job_1789023718167_0001
2026-09-10 13:32:41,360 INFO mapreduce.Job: Job job_1789023718167_0001 running in uber mode : false
2026-09-10 13:32:41,362 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 13:32:49,525 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 13:33:06,842 INFO mapreduce.Job: map 100% reduce 11%
2026-09-10 13:33:12,968 INFO mapreduce.Job: map 100% reduce 22%
2026-09-10 13:33:26,218 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 13:33:27,251 INFO mapreduce.Job: Job job_1789023718167_0001 completed successfully
2026-09-10 13:33:27,436 INFO mapreduce.Job: Counters: 54
File System Counters
FILE: Number of bytes read=11576959
FILE: Number of bytes written=24399747
FILE: Number of read operations=0
FILE: Number of large read operations=0
FILE: Number of write operations=0
HDFS: Number of bytes read=345222759
HDFS: Number of bytes written=216
HDFS: Number of read operations=14
```

Output

```
Administrator: Command Prompt

Map output records=936811
Map output bytes=9703331
Map output materialized bytes=11576971
Input split bytes=378
Combine input records=0
Combine output records=0
Reduce input groups=12
Reduce shuffle bytes=11576971
Reduce input records=936811
Reduce output records=12
Spilled Records=1873622
Shuffled Maps =3
Failed Shuffles=2
Merged Map outputs=3
GC time elapsed (ms)=395
CPU time spent (ms)=10146
Physical memory (bytes) snapshot=1713971200
Virtual memory (bytes) snapshot=2469593088
Total committed heap usage (bytes)=1678245888
Peak Map Physical memory (bytes)=514572288
Peak Map Virtual memory (bytes)=734294016
Peak Reduce Physical memory (bytes)=249040896
Peak Reduce Virtual memory (bytes)=419868672

Shuffle Errors
BAD_ID=2
CONNECTION=0
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0

File Input Format Counters
Bytes Read=345222381
File Output Format Counters
Bytes Written=216

C:\BDAL\BDAPProject>hdfs dfs -ls /chicago/output/weather_wise_crash
Found 2 items
-rw-r--r-- 1 Biswas supergroup 0 2026-09-10 13:33 /chicago/output/weather_wise_crash/_SUCCESS
-rw-r--r-- 1 Biswas supergroup 216 2026-09-10 13:33 /chicago/output/weather_wise_crash/part-r-00000
```

```

C:\BDAL\BDAPProject>hdfs dfs -cat /chicago/output/weather_wise_crash/part-r-00000
BLOWING SAND; SOIL; DIRT          9
BLOWING SNOW      487
CLEAR      734431
CLOUDY/OVERCAST 27600
FOG/SMOKE/HAZE   1396
FREEZING RAIN/DRIZZLE 2309
OTHER      3030
RAIN       80851
SEVERE CROSS WIND GATE 159
SLEET/HAIL    1111
SNOW        30710
UNKNOWN 54718
C:\BDAL\BDAPProject>

```

6.2 Road Surface-Wise Total Injuries

Road Surface-Wise Total Injuries: This analysis calculates the total number of injuries recorded under each road surface condition.

This analysis comprises of:

1. **Mapper Class:** Extracts the road surface condition and total injury value from every valid crash record.
2. **Reducer Class:** Adds all injury values for the same road surface condition.
3. **Driver Class:** Configures and executes the MapReduce job.

Mapper Class

```

public static class InjuryMapper
    extends Mapper<Object, Text, Text, IntWritable> {

    private Text surface = new Text();
    private IntWritable injuryValue = new IntWritable();

    public void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Skip header
        if (line.startsWith("CRASH_RECORD_ID")) {
            return;
        }

        String[] fields = line.split(",", -1);

        if (fields.length != 24) {
            return;
        }

```

```

        String roadSurface = fields[9].trim();
        String injuryText = fields[18].trim();

        // Skip blank values
        if (roadSurface.isEmpty() || injuryText.isEmpty()) {
            return;
        }

        try {
            int injuries = (int) Double.parseDouble(injuryText);

            surface.set(roadSurface);
            injuryValue.set(injuries);

            context.write(surface, injuryValue);

        } catch (NumberFormatException e) {
            // Skip invalid numeric rows
        }
    }
}

```

Reducer Class

```
public static class InjuryReducer
    extends Reducer<Text, IntWritable, Text, IntWritable> {

    private IntWritable result = new IntWritable();

    public void reduce(Text key,
                      Iterable<IntWritable> values,
                      Context context)
        throws IOException, InterruptedException {

        int totalInjuries = 0;

        for (IntWritable val : values) {
            totalInjuries += val.get();
        }

        result.set(totalInjuries);
        context.write(key, result);
    }
}
```

Driver Class

```
public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    Job job = Job.getInstance(
        conf,
        "Road Surface Wise Total Injury Analysis"
    );

    job.setJarByClass(RoadSurfaceInjury.class);
    job.setMapperClass(InjuryMapper.class);
    job.setReducerClass(InjuryReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(IntWritable.class);

    FileInputFormat.addInputPath(
        job,
        new Path(args[0])
    );

    FileOutputFormat.setOutputPath(
        job,
        new Path(args[1])
    );

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
```

Execution

```
Administrator: Command Prompt

C:\BDAL\BDAPProject>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*;%HADOOP_HOME%\share\hadoop\mapreduce\*" -d . RoadSurfaceInjury.java

C:\BDAL\BDAPProject>jar -cvf RoadSurfaceInjury.jar RoadSurfaceInjury.class RoadSurfaceInjury$InjuryMapper.class RoadSurfaceInjury$InjuryReducer.class
added manifest
adding: RoadSurfaceInjury.class(in = 1507) (out= 816)(deflated 45%)
adding: RoadSurfaceInjury$InjuryMapper.class(in = 2110) (out= 958)(deflated 54%)
adding: RoadSurfaceInjury$InjuryReducer.class(in = 1763) (out= 745)(deflated 57%)

C:\BDAL\BDAPProject>hadoop jar RoadSurfaceInjury.jar RoadSurfaceInjury /chicago/input/ChicagoTrafficCrashes.csv /chicago/output/road_surface_injury
2026-09-10 13:40:04,316 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 13:40:05,342 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with
ToolRunner to remedy this.
2026-09-10 13:40:05,370 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Biswas/.staging/job_1789023718167_0002
2026-09-10 13:40:05,811 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 13:40:05,930 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 13:40:06,085 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789023718167_0002
2026-09-10 13:40:06,085 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 13:40:06,375 INFO conf.Configuration: resource-types.xml not found
2026-09-10 13:40:06,376 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-10 13:40:06,535 INFO impl.YarnClientImpl: Submitted application application_1789023718167_0002
2026-09-10 13:40:06,594 INFO mapreduce.Job: The url to track the job: http://DESKTOP-3CQRA5T:8088/proxy/application_1789023718167_0002/
2026-09-10 13:40:06,596 INFO mapreduce.Job: Running job: job_1789023718167_0002
2026-09-10 13:40:15,899 INFO mapreduce.Job: Job job_1789023718167_0002 running in uber mode : false
2026-09-10 13:40:15,901 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 13:40:24,095 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 13:40:42,430 INFO mapreduce.Job: map 100% reduce 22%
2026-09-10 13:40:45,476 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 13:40:46,511 INFO mapreduce.Job: Job job_1789023718167_0002 completed successfully
2026-09-10 13:40:46,758 INFO mapreduce.Job: Counters: 54

File System Counters
  FILE: Number of bytes read=9998336
  FILE: Number of bytes written=21242529
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=345222759
  HDFS: Number of bytes written=91
  HDFS: Number of read operations=14
  HDFS: Number of large read operations=0
```

Output

```
Reduce output records=7
Spilled Records=1869530
Shuffled Maps =3
Failed Shuffles=1
Merged Map outputs=3
GC time elapsed (ms)=428
CPU time spent (ms)=10354
Physical memory (bytes) snapshot=1677180928
Virtual memory (bytes) snapshot=2396819456
Total committed heap usage (bytes)=1704984576
Peak Map Physical memory (bytes)=511504384
Peak Map Virtual memory (bytes)=669560832
Peak Reduce Physical memory (bytes)=249376768
Peak Reduce Virtual memory (bytes)=417726464

Shuffle Errors
  BAD_ID=1
  CONNECTION=0
  IO_ERROR=0
  WRONG_LENGTH=0
  WRONG_MAP=0
  WRONG_REDUCE=0

File Input Format Counters
  Bytes Read=345222381
File Output Format Counters
  Bytes Written=91

C:\BDAL\BDAPProject>hdfs dfs -ls /chicago/output/road_surface_injury
Found 2 items
-rw-r--r-- 1 Biswas supergroup          0 2026-09-10 13:40 /chicago/output/road_surface_injury/_SUCCESS
-rw-r--r-- 1 Biswas supergroup        91 2026-09-10 13:40 /chicago/output/road_surface_injury/part-r-00000
```

```
C:\BDAL\BDAPProject>hdfs dfs -cat /chicago/output/road_surface_injury/part-r-00000
DRY      139394
ICE       1070
OTHER     520
SAND; MUD; DIRT  57
SNOW OR SLUSH  4311
UNKNOWN  7692
WET      29300
```

6.3 Crash Type-Wise Average Speed

Crash Type-Wise Average Speed: This analysis calculates the average posted speed limit for each crash type.

This analysis comprises of:

1. **Mapper Class:** Extracts the crash type and posted speed limit.
2. **Reducer Class:** Calculates the total speed and number of records, then finds the average speed.
3. **Driver Class:** Configures the MapReduce job.

Mapper Class

```
public static class SpeedMapper
    extends Mapper<Object, Text, Text, Text> {

    public void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Skip header
        if (line.startsWith("CRASH_RECORD_ID")) {
            return;
        }

        String[] fields = line.split(",", -1);

        if (fields.length != 24) {
            return;
        }

        String crashType = fields[7].trim();
        String speedText = fields[2].trim();

        if (crashType.isEmpty() || speedText.isEmpty()) {
            return;
        }

        try {
            double speed = Double.parseDouble(speedText);

            context.write(
                new Text(crashType),
                new Text(speed + ",1")
            );
        } catch (NumberFormatException e) {
            // Skip invalid speed values
        }
    }
}
```

Reducer Class

```
public static class SpeedReducer
    extends Reducer<Text, Text, Text, Text> {

    public void reduce(Text key,
        Iterable<Text> values,
        Context context)
        throws IOException, InterruptedException {

        double totalSpeed = 0;
        int count = 0;

        for (Text value : values) {

            String[] parts = value.toString().split(",");

            double speed = Double.parseDouble(parts[0]);
            int recordCount = Integer.parseInt(parts[1]);

            totalSpeed += speed;
            count += recordCount;
        }

        if (count > 0) {

            double averageSpeed = totalSpeed / count;

            context.write(
                key,
                new Text(String.format("%.2f", averageSpeed))
            );
        }
    }
}
```

Driver Class

```
public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    Job job = Job.getInstance(
        conf,
        "Crash Type Wise Average Speed"
    );

    job.setJarByClass(CrashTypeAverageSpeed.class);
    job.setMapperClass(SpeedMapper.class);
    job.setReducerClass(SpeedReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(Text.class);

    FileInputFormat.addInputPath(
        job,
        new Path(args[0])
    );

    FileOutputFormat.setOutputPath(
        job,
        new Path(args[1])
    );

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
```

Execution

```
Administrator: Command Prompt

C:\BDAL\BDAPProject>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*; %HADOOP_HOME%\share\hadoop\mapreduce\*" -d . CrashTypeAverageSpeed.java

C:\BDAL\BDAPProject>jar -cvf CrashTypeAverageSpeed.jar CrashTypeAverageSpeed.class CrashTypeAverageSpeed$SpeedMapper.class CrashTypeAverageSpeed$SpeedReducer.class
added manifest
adding: CrashTypeAverageSpeed.class(in = 1471) (out= 790)(deflated 46%)
adding: CrashTypeAverageSpeed$SpeedMapper.class(in = 2033) (out= 915)(deflated 54%)
adding: CrashTypeAverageSpeed$SpeedReducer.class(in = 2126) (out= 927)(deflated 56%)

C:\BDAL\BDAPProject>hadoop jar CrashTypeAverageSpeed.jar CrashTypeAverageSpeed /chicago/input/ChicagoTrafficCrashes.csv /chicago/output/crash_type_avg_speed
2026-09-10 14:01:49,429 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 14:01:50,439 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with
ToolRunner to remedy this.
2026-09-10 14:01:50,460 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Biswas/.staging/job_1789023718167_0003
2026-09-10 14:01:50,896 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 14:01:50,992 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 14:01:51,147 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789023718167_0003
2026-09-10 14:01:51,147 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 14:01:51,442 INFO conf.Configuration: resource-types.xml not found
2026-09-10 14:01:51,444 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-10 14:01:51,610 INFO impl.YarnClientImpl: Submitted application application_1789023718167_0003
2026-09-10 14:01:51,683 INFO mapreduce.Job: The url to track the job: http://DESKTOP-JCQRAST:8088/proxy/application_1789023718167_0003/
2026-09-10 14:01:51,685 INFO mapreduce.Job: Running job: job_1789023718167_0003
2026-09-10 14:01:58,915 INFO mapreduce.Job: Job job_1789023718167_0003 running in uber mode : false
2026-09-10 14:01:58,919 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 14:02:07,351 INFO mapreduce.Job: map 33% reduce 0%
2026-09-10 14:02:08,388 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 14:02:25,698 INFO mapreduce.Job: map 100% reduce 11%
2026-09-10 14:02:31,777 INFO mapreduce.Job: map 100% reduce 22%
2026-09-10 14:02:46,015 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 14:02:46,206 INFO mapreduce.Job: Job job_1789023718167_0003 completed successfully
2026-09-10 14:02:46,529 INFO mapreduce.Job: Counters: 54

File System Counters
  FILE: Number of bytes read=21989762
  FILE: Number of bytes written=45225341
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=345222759
```

Output

```
File Input Format Counters
  Bytes Read=345222381
File Output Format Counters
  Bytes Written=347

C:\BDAL\BDAPProject>hdfs dfs -ls /chicago/output/crash_type_avg_speed
Found 2 items
-rw-r--r--  1 Biswas supergroup          0 2026-09-10 14:02 /chicago/output/crash_type_avg_speed/_SUCCESS
-rw-r--r--  1 Biswas supergroup    347 2026-09-10 14:02 /chicago/output/crash_type_avg_speed/part-r-00000

C:\BDAL\BDAPProject>hdfs dfs -cat /chicago/output/crash_type_avg_speed/part-r-00000
ANGLE      28.34
ANIMAL     29.62
FIXED OBJECT 27.19
HEAD ON    28.29
OTHER NONCOLLISION 29.69
OTHER OBJECT 27.07
OVERTURNED 30.23
PARKED MOTOR VEHICLE 26.10
PEDALCYCLIST 29.01
PEDESTRIAN 28.27
REAR END   29.82
REAR TO FRONT 26.38
REAR TO REAR 20.19
REAR TO SIDE 23.08
SIDESWIPE OPPOSITE DIRECTION 27.35
SIDESWIPE SAME DIRECTION 29.77
TRAIN      28.92
TURNING    29.65
```

6.4 Road Surface-Wise Injury Rate

Road Surface-Wise Injury Rate: This analysis calculates the percentage of crashes involving at least one injury for every road surface condition.

This analysis comprises of:

1. **Mapper Class:** The InjuryRateMapper extracts the road surface condition and total injury value.
2. **Reducer Class:** The InjuryRateReducer adds the number of injury crashes and total crashes for each road surface.
3. **Driver Class:** The Driver Class creates the Road Surface Wise Injury Rate job and connects the Mapper and Reducer.

Mapper Class

```
public static class InjuryRateMapper
    extends Mapper<Object, Text, Text, Text> {

    public void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Skip header
        if (line.startsWith("CRASH_RECORD_ID")) {
            return;
        }

        String[] fields = line.split(",", -1);

        if (fields.length != 24) {
            return;
        }
        // Column 9 = ROADWAY_SURFACE_COND
        String roadSurface = fields[9].trim();

        // Column 18 = INJURIES_TOTAL
        String injuryText = fields[18].trim();

        if (roadSurface.isEmpty() || injuryText.isEmpty()) {
            return;
        }

        try {
            double injuries = Double.parseDouble(injuryText)

            // injuryCrash,totalCrash
            if (injuries > 0) {
                context.write(
                    new Text(roadSurface),
                    new Text("1,1")
                );
            } else {
                context.write(
                    new Text(roadSurface),
                    new Text("0,1")
                );
            }
        } catch (NumberFormatException e) {
            // Skip invalid rows
        }
    }
}
```

Reducer Class

```
public static class InjuryRateReducer
    extends Reducer<Text, Text, Text, Text> {

    public void reduce(Text key,
        Iterable<Text> values,
        Context context)
        throws IOException, InterruptedException {

        int injuryCrashes = 0;
        int totalCrashes = 0;

        for (Text value : values) {
            String[] parts = value.toString().split(",");
            injuryCrashes += Integer.parseInt(parts[0]);
            totalCrashes += Integer.parseInt(parts[1]);
        }

        if (totalCrashes > 0) {

            double injuryRate =
                ((double) injuryCrashes / totalCrashes) * 100.0;

            String result =
                String.format("%.2f%%", injuryRate);

            context.write(key, new Text(result));
        }
    }
}
```

Driver Class

```
public static void main(String[] args) throws Exception {

    Configuration conf = new Configuration();

    Job job = Job.getInstance(
        conf,
        "Road Surface Wise Injury Rate"
    );

    job.setJarByClass(RoadSurfaceInjuryRate.class);
    job.setMapperClass(InjuryRateMapper.class);
    job.setReducerClass(InjuryRateReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(Text.class);

    FileInputFormat.addInputPath(
        job,
        new Path(args[0])
    );

    FileOutputFormat.setOutputPath(
        job,
        new Path(args[1])
    );

    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
```

Execution

```
Administrator: Command Prompt

C:\BDAL\BDAPProject>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*; %HADOOP_HOME%\share\hadoop\mapreduce\*" -d . RoadSurfaceInjuryRate.java

C:\BDAL\BDAPProject>jar -cvf RoadSurfaceInjuryRate.jar RoadSurfaceInjuryRate.class RoadSurfaceInjuryRate$InjuryRateMapper.class RoadSurfaceInjuryRate$InjuryRateReducer.class

added manifest
adding: RoadSurfaceInjuryRate.class(in = 1491) (out= 792)(deflated 46%)
adding: RoadSurfaceInjuryRate$InjuryRateMapper.class(in = 1936) (out= 883)(deflated 54%)
adding: RoadSurfaceInjuryRate$InjuryRateReducer.class(in = 2092) (out= 917)(deflated 56%)

C:\BDAL\BDAPProject>hadoop jar RoadSurfaceInjuryRate.jar RoadSurfaceInjuryRate /chicago/input/ChicagoTrafficCrashes.csv /chicago/output/road_surface_injury_rate
2026-09-10 14:12:41,565 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 14:12:42,607 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with
ToolRunner to remedy this.
2026-09-10 14:12:42,634 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Biswas/.staging/job_1789023718167_0004
2026-09-10 14:12:43,071 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 14:12:43,587 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 14:12:43,789 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789023718167_0004
2026-09-10 14:12:43,789 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 14:12:44,121 INFO conf.Configuration: resource-types.xml not found
2026-09-10 14:12:44,122 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-10 14:12:44,311 INFO impl.YarnClientImpl: Submitted application application_1789023718167_0004
2026-09-10 14:12:44,386 INFO mapreduce.Job: The url to track the job: http://DESKTOP-3CQRAST:8088/proxy/application_1789023718167_0004/
2026-09-10 14:12:44,388 INFO mapreduce.Job: Running job: job_1789023718167_0004
2026-09-10 14:12:53,641 INFO mapreduce.Job: Job job_1789023718167_0004 running in uber mode : false
2026-09-10 14:12:53,644 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 14:13:01,082 INFO mapreduce.Job: map 67% reduce 0%
2026-09-10 14:13:02,112 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 14:13:18,340 INFO mapreduce.Job: map 100% reduce 11%
2026-09-10 14:13:24,452 INFO mapreduce.Job: map 100% reduce 22%
2026-09-10 14:13:38,716 INFO mapreduce.Job: map 100% reduce 100%
2026-09-10 14:13:38,745 INFO mapreduce.Job: Job job_1789023718167_0004 completed successfully
2026-09-10 14:13:38,931 INFO mapreduce.Job: Counters: 54
File System Counters
  FILE: Number of bytes read=9998336
  FILE: Number of bytes written=21242545
  FILE: Number of read operations=0
  FILE: Number of large read operations=0
  FILE: Number of write operations=0
  HDFS: Number of bytes read=345222759
```

Output

```
Bytes Written=104
C:\BDAL\BDAPProject>hdfs dfs -ls /chicago/output/road_surface_injury_rate
Found 2 items
-rw-r--r--  1 Biswas supergroup          0 2026-09-10 14:13 /chicago/output/road_surface_injury_rate/_SUCCESS
-rw-r--r--  1 Biswas supergroup       104 2026-09-10 14:13 /chicago/output/road_surface_injury_rate/part-r-00000

C:\BDAL\BDAPProject>hdfs dfs -cat /chicago/output/road_surface_injury_rate/part-r-00000
DRY      14.59%
ICE      12.02%
OTHER    15.73%
SAND; MUD; DIRT 14.67%
SNOW OR SLUSH  10.79%
UNKNOWN  7.13%
WET      17.37%

C:\BDAL\BDAPProject>
```

6.5 High-Risk Condition Score

High-Risk Condition Score: This analysis combines weather, lighting and road surface conditions to measure the average severity of crashes under different environmental conditions.

This analysis comprises of:

1. **Mapper Class:** Extracts environmental conditions and injury information.
2. **Reducer Class:** Calculates a weighted severity score for every combination.
3. **Driver Class:** Configures and runs the MapReduce job.

Mapper Class

```
public static class RiskMapper
    extends Mapper<Object, Text, Text, Text> {

    public void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Skip header
        if (line.startsWith("CRASH_RECORD_ID")) {
            return;
        }

        String[] fields = line.split(",", -1);

        if (fields.length != 24) {
            return;
        }

        String weather = fields[5].trim();
        String lighting = fields[6].trim();
        String roadSurface = fields[9].trim();

        String injuriesText = fields[18].trim();

        String fatalText = fields[19].trim();
        String incapacitatingText = fields[20].trim();

        // Skip rows with missing required values
        if (weather.isEmpty() ||
            lighting.isEmpty() ||
            roadSurface.isEmpty() ||
            injuriesText.isEmpty() ||
            fatalText.isEmpty() ||
            incapacitatingText.isEmpty()) {
            return;
        }

        try {

            double injuries =
                Double.parseDouble(injuriesText);

            double fatal =
                Double.parseDouble(fatalText);

            double incapacitating =
                Double.parseDouble(incapacitatingText);
```

```

        // Combination key
        String condition =
            weather + " | " +
            lighting + " | " +
            roadSurface;

        /*
        Values:total injuries,incapacitating injur
        */
        String outputValue =
            injuries + "," +
            incapacitating + "," +
            fatal + ",1";

        context.write(
            new Text(condition),
            new Text(outputValue)
        );

    } catch (NumberFormatException e) {
        // Skip invalid numeric records
    }
}
}

```

Reducer Class

```

public static class RiskReducer
    extends Reducer<Text, Text, Text, Text> {

    public void reduce(Text key,
        Iterable<Text> values,
        Context context)
        throws IOException, InterruptedException {

        double totalInjuries = 0;
        double totalIncapacitating = 0;
        double totalFatal = 0;

        int totalCrashes = 0;

        for (Text value : values) {

            String[] parts =
                value.toString().split(",");

            totalInjuries +=
                Double.parseDouble(parts[0]);

            totalIncapacitating +=
                Double.parseDouble(parts[1]);

            totalFatal +=
                Double.parseDouble(parts[2]);

            totalCrashes +=
                Integer.parseInt(parts[3]);
        }

        if (totalCrashes > 0) {

            double riskScore =
                (totalInjuries
                + (2 * totalIncapacitating)
                + (5 * totalFatal))
                / totalCrashes;

            String result =
                String.format(
                    Locale.US,
                    "%.4f",
                    riskScore
                );

            context.write(
                key,
                new Text(result)
            );
        }
    }
}

```

Driver Class

```
public static void main(String[] args)
    throws Exception {

    Configuration conf = new Configuration();

    Job job =
        Job.getInstance(
            conf,
            "High Risk Condition Score"
        );

    job.setJarByClass(HighRiskConditionScore.class);
    job.setMapperClass(RiskMapper.class);
    job.setReducerClass(RiskReducer.class);
    job.setOutputKeyClass(Text.class);
    job.setOutputValueClass(Text.class);
    FileInputFormat.addInputPath(
        job,
        new Path(args[0])
    );
    FileOutputFormat.setOutputPath(
        job,
        new Path(args[1])
    );
    System.exit(job.waitForCompletion(true) ? 0 : 1);
}
```

Execution

```
Administrator: Command Prompt
C:\BDAL\BDAPProject>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*; %HADOOP_HOME%\share\hadoop\mapreduce\*" -d . HighRiskConditionScore.java
C:\BDAL\BDAPProject>jar -cvf HighRiskConditionScore.jar HighRiskConditionScore.class HighRiskConditionScore$RiskMapper.class HighRiskConditionScore$RiskReducer.class
Added manifest
Adding: HighRiskConditionScore.class(in = 1479) (out= 802)(deflated 45%)
Adding: HighRiskConditionScore$RiskMapper.class(in = 2321) (out= 1072)(deflated 53%)
Adding: HighRiskConditionScore$RiskReducer.class(in = 2326) (out= 1044)(deflated 55%)
C:\BDAL\BDAPProject>hadoop jar HighRiskConditionScore.jar HighRiskConditionScore /chicago/input/ChicagoTrafficCrashes.csv /chicago/output/high_risk_condition_score
2026-09-10 14:17:30,817 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 14:17:31,869 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with
ToolRunner to remedy this.
2026-09-10 14:17:31,891 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Biswas/.staging/job_1789023718167_0005
2026-09-10 14:17:32,317 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 14:17:32,406 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 14:17:32,554 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789023718167_0005
2026-09-10 14:17:32,554 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 14:17:32,814 INFO conf.Configuration: resource-types.xml not found
2026-09-10 14:17:32,815 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-10 14:17:32,973 INFO impl.YarnClientImpl: Submitted application application_1789023718167_0005
2026-09-10 14:17:33,052 INFO mapreduce.Job: The url to track the job: http://DESKTOP-3CQRA5T:8088/proxy/application_1789023718167_0005/
2026-09-10 14:17:33,054 INFO mapreduce.Job: Running job: job_1789023718167_0005
2026-09-10 14:17:41,286 INFO mapreduce.Job: Job job_1789023718167_0005 running in uber mode : false
2026-09-10 14:17:41,289 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 14:17:50,538 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 14:17:54,618 INFO mapreduce.Job: Task Id : attempt_1789023718167_0005_r_000000_0, Status : FAILED
Error: org.apache.hadoop.mapreduce.task.reduce.Shuffle$ShuffleError: error in shuffle in fetcher#1
    at org.apache.hadoop.mapreduce.task.reduce.Shuffle.run(Shuffle.java:136)
    at org.apache.hadoop.mapred.ReduceTask.run(ReduceTask.java:377)
    at org.apache.hadoop.mapred.YarnChild$2.run(YarnChild.java:178)
    at java.security.AccessController.doPrivileged(Native Method)
    at javax.security.auth.Subject.doAs(Subject.java:422)
    at org.apache.hadoop.security.authentication.util.SubjectUtil.doAs(SubjectUtil.java:247)
    at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1951)
    at org.apache.hadoop.mapred.YarnChild.main(YarnChild.java:172)
Caused by: java.io.IOException: Exceeded MAX_FAILED_UNIQUE_FETCHES; bailing-out.
    at org.apache.hadoop.mapreduce.task.reduce.ShuffleSchedulerImpl.checkReducerHealth(ShuffleSchedulerImpl.java:396)
    at org.apache.hadoop.mapreduce.task.reduce.ShuffleSchedulerImpl.copyFailed(ShuffleSchedulerImpl.java:311)
    at org.apache.hadoop.mapreduce.task.reduce.Fetcher.copyFromHost(Fetcher.java:365)
    at org.apache.hadoop.mapreduce.task.reduce.Fetcher.run(Fetcher.java:203)
```

Output

```
Administrator: Command Prompt
C:\BDAL\BDAPProject>hdfs dfs -ls /chicago/output/high_risk_condition_score
Found 2 items
-rw-r--r-- 1 Biswas supergroup          0 2026-09-10 14:18 /chicago/output/high_risk_condition_score/_SUCCESS
-rw-r--r-- 1 Biswas supergroup    15518 2026-09-10 14:18 /chicago/output/high_risk_condition_score/part-r-00000

C:\BDAL\BDAPProject>hdfs dfs -cat /chicago/output/high_risk_condition_score/part-r-00000
BLOWING SAND; SOIL; DIRT | DARKNESS | DRY      0.0000
BLOWING SAND; SOIL; DIRT | DARKNESS | SAND; MUD; DIRT 2.0000
BLOWING SAND; SOIL; DIRT | DARKNESS; LIGHTED ROAD | DRY 0.0000
BLOWING SAND; SOIL; DIRT | DARKNESS; LIGHTED ROAD | UNKNOWN 0.0000
BLOWING SAND; SOIL; DIRT | DAYLIGHT | DRY      0.0000
BLOWING SAND; SOIL; DIRT | DAYLIGHT | UNKNOWN 0.0000
BLOWING SAND; SOIL; DIRT | UNKNOWN | UNKNOWN 0.0000
BLOWING SNOW | DARKNESS | ICE      0.0000
BLOWING SNOW | DARKNESS | SNOW OR SLUSH 0.3333
BLOWING SNOW | DARKNESS | UNKNOWN 0.0000
BLOWING SNOW | DARKNESS | WET      0.2222
BLOWING SNOW | DARKNESS; LIGHTED ROAD | DRY      0.4000
BLOWING SNOW | DARKNESS; LIGHTED ROAD | ICE      1.0000
BLOWING SNOW | DARKNESS; LIGHTED ROAD | SNOW OR SLUSH 0.2815
BLOWING SNOW | DARKNESS; LIGHTED ROAD | UNKNOWN 0.3333
BLOWING SNOW | DARKNESS; LIGHTED ROAD | WET      0.3725
BLOWING SNOW | DAWN | SNOW OR SLUSH 0.0000
BLOWING SNOW | DAYLIGHT | DRY      0.1000
BLOWING SNOW | DAYLIGHT | ICE      0.0667
BLOWING SNOW | DAYLIGHT | OTHER 0.0000
BLOWING SNOW | DAYLIGHT | SNOW OR SLUSH 0.2982
BLOWING SNOW | DAYLIGHT | UNKNOWN 0.0000
BLOWING SNOW | DAYLIGHT | WET      0.3455
BLOWING SNOW | DUSK | SNOW OR SLUSH 0.3571
BLOWING SNOW | DUSK | UNKNOWN 0.0000
BLOWING SNOW | DUSK | WET      0.8571
BLOWING SNOW | UNKNOWN | SNOW OR SLUSH 0.0000
BLOWING SNOW | UNKNOWN | UNKNOWN 0.0000
CLEAR | DARKNESS | DRY      0.2425
CLEAR | DARKNESS | ICE      0.2228
CLEAR | DARKNESS | OTHER 0.2692
CLEAR | DARKNESS | SAND; MUD; DIRT 0.3810
CLEAR | DARKNESS | SNOW OR SLUSH 0.1404
```

6.6 Traffic Control-Wise Severe Crash Rate

Traffic Control-Wise Severe Crash Rate: This analysis calculates the percentage of severe crashes for every type of traffic control device.

This analysis comprises of:

1. **Mapper Class:** Reads each crash record and identifies whether it is severe based on fatal or incapacitating injuries for each traffic control type.
2. **Reducer Class:** Calculates the total number of severe crashes and the total number of valid crashes for each traffic control category.
3. **Driver Class:** Creates the Traffic Control Wise Severe Crash Rate job and configures the input, output, Mapper and Reducer.

Mapper Class

```
public static class SevereCrashMapper
    extends Mapper<Object, Text, Text, Text> {

    public void map(Object key, Text value, Context context)
        throws IOException, InterruptedException {

        String line = value.toString();

        // Skip header
        if (line.startsWith("CRASH_RECORD_ID")) {
            return;
        }

        String[] fields = line.split(",", -1);

        if (fields.length != 24) {
            return;
        }

        String trafficControl = fields[3].trim();
        String fatalText = fields[19].trim();
        String incapacitatingText = fields[20].trim();

        // Skip missing values
        if (trafficControl.isEmpty() || fatalText.isEmpty() || incapacitatingText.isEmpty()) {
            return;
        }
    }
}
```

```
try {

    double fatal =
        Double.parseDouble(fatalText);

    double incapacitating =
        Double.parseDouble(incapacitatingText);

    // severeCrash, totalCrash
    if (fatal > 0 || incapacitating > 0) {

        context.write(
            new Text(trafficControl),
            new Text("1,1")
        );

    } else {

        context.write(
            new Text(trafficControl),
            new Text("0,1")
        );

    }

} catch (NumberFormatException e) {
    // Skip invalid rows
}
```

Reducer Class

```
public static class SevereCrashReducer
    extends Reducer<Text, Text, Text, Text> {

    public void reduce(Text key,
        Iterable<Text> values,
        Context context)
        throws IOException, InterruptedException {

        int severeCrashes = 0;
        int totalCrashes = 0;

        for (Text value : values) {

            String[] parts =
                value.toString().split(",");

            severeCrashes +=
                Integer.parseInt(parts[0]);

            totalCrashes +=
                Integer.parseInt(parts[1]);

        }
    }
}
```

```
if (totalCrashes > 0) {

    double severeCrashRate =
        ((double) severeCrashes
        / totalCrashes) * 100.0;

    String result =
        String.format(
            Locale.US,
            "%.2f%%",
            severeCrashRate
        );

    context.write(
        key,
        new Text(result)
    );

}
}
```

Driver Class

```
public static void main(String[] args)
    throws Exception {

    Configuration conf =
        new Configuration();

    Job job =
        Job.getInstance(
            conf,
            "Traffic Control Wise Severe Crash Rate"
        );

    job.setJarByClass(
        TrafficControlSevereCrashRate.class
    );

    job.setMapperClass(
        SevereCrashMapper.class
    );

    job.setReducerClass(
        SevereCrashReducer.class
    );

    job.setOutputKeyClass(
        Text.class
    );

    job.setOutputValueClass(
        Text.class
    );

    FileInputFormat.addInputPath(
        job,
        new Path(args[0])
    );

    FileOutputFormat.setOutputPath(
        job,
        new Path(args[1])
    );

    System.exit(
        job.waitForCompletion(true)
            ? 0
            : 1
    );
}
```

Execution

```
Administrator: Command Prompt

C:\BDAL\BDAPProject>javac -classpath "%HADOOP_HOME%\share\hadoop\common\*;%;HADOOP_HOME%\share\hadoop\mapreduce\*" -d . TrafficControlSevereCrashRate.java

C:\BDAL\BDAPProject>jar -cvf TrafficControlSevereCrashRate.jar TrafficControlSevereCrashRate.class TrafficControlSevereCrashRate$SevereCrashMapper.class TrafficControlSevereCrashRate$SevereCrashReducer.class
added manifest
adding: TrafficControlSevereCrashRate.class(in = 1548) (out= 822)(deflated 46%)
adding: TrafficControlSevereCrashRate$SevereCrashMapper.class(in = 2022) (out= 930)(deflated 54%)
adding: TrafficControlSevereCrashRate$SevereCrashReducer.class(in = 2215) (out= 964)(deflated 56%)

C:\BDAL\BDAPProject>hadoop jar TrafficControlSevereCrashRate.jar TrafficControlSevereCrashRate /chicago/input/ChicagoTrafficCrashes.csv /chicago/output/traffic_control_severe_crash_rate
2026-09-10 14:30:22,144 INFO client.DefaultNoHARMFailoverProxyProvider: Connecting to ResourceManager at /0.0.0.0:8032
2026-09-10 14:30:23,178 WARN mapreduce.JobResourceUploader: Hadoop command-line option parsing not performed. Implement the Tool interface and execute your application with
ToolRunner to remedy this.
2026-09-10 14:30:23,200 INFO mapreduce.JobResourceUploader: Disabling Erasure Coding for path: /tmp/hadoop-yarn/staging/Biswas/.staging/job_1789023718167_0006
2026-09-10 14:30:23,655 INFO input.FileInputFormat: Total input files to process : 1
2026-09-10 14:30:23,993 INFO mapreduce.JobSubmitter: number of splits:3
2026-09-10 14:30:24,172 INFO mapreduce.JobSubmitter: Submitting tokens for job: job_1789023718167_0006
2026-09-10 14:30:24,172 INFO mapreduce.JobSubmitter: Executing with tokens: []
2026-09-10 14:30:24,428 INFO conf.Configuration: resource-types.xml not found
2026-09-10 14:30:24,429 INFO resource.ResourceUtils: Unable to find 'resource-types.xml'.
2026-09-10 14:30:24,988 INFO impl.YarnClientImpl: Submitted application application_1789023718167_0006
2026-09-10 14:30:24,982 INFO mapreduce.Job: The url to track the job: http://DESKTOP-3CQRAST:8088/proxy/application_1789023718167_0006/
2026-09-10 14:30:24,989 INFO mapreduce.Job: Running job: job_1789023718167_0006
2026-09-10 14:30:33,288 INFO mapreduce.Job: Job job_1789023718167_0006 running in uber mode : false
2026-09-10 14:30:33,294 INFO mapreduce.Job: map 0% reduce 0%
2026-09-10 14:30:42,764 INFO mapreduce.Job: map 33% reduce 0%
2026-09-10 14:30:43,776 INFO mapreduce.Job: map 100% reduce 0%
2026-09-10 14:30:49,876 INFO mapreduce.Job: Task Id : attempt_1789023718167_0006_r_000000_0, Status : FAILED
Error: org.apache.hadoop.mapreduce.task.reduce.Shuffle$ShuffleError: error in shuffle in fetcher#1
    at org.apache.hadoop.mapreduce.task.reduce.Shuffle.run(Shuffle.java:136)
    at org.apache.hadoop.mapred.ReduceTask.run(ReduceTask.java:377)
    at org.apache.hadoop.mapred.YarnChild$2.run(YarnChild.java:178)
    at java.security.AccessController.doPrivileged(Native Method)
    at javax.security.auth.Subject.doAs(Subject.java:422)
    at org.apache.hadoop.security.authentication.util.SubjectUtil.doAs(SubjectUtil.java:247)
    at org.apache.hadoop.security.UserGroupInformation.doAs(UserGroupInformation.java:1951)
    at org.apache.hadoop.mapred.YarnChild.main(YarnChild.java:172)
```

Output

```
IO_ERROR=0
WRONG_LENGTH=0
WRONG_MAP=0
WRONG_REDUCE=0
File Input Format Counters
  Bytes Read=345222381
File Output Format Counters
  Bytes Written=416

C:\BDAL\BDAPProject>hdfs dfs -ls /chicago/output/traffic_control_severe_rate
Found 2 items
-rw-r--r--  1 Biswas supergroup          0 2026-09-10 14:31 /chicago/output/traffic_control_severe_rate/_SUCCESS
-rw-r--r--  1 Biswas supergroup      416 2026-09-10 14:31 /chicago/output/traffic_control_severe_rate/part-r-000000

C:\BDAL\BDAPProject>hdfs dfs -cat /chicago/output/traffic_control_severe_rate/part-r-000000
BICYCLE CROSSING SIGN    5.71%
DELINEATORS             3.94%
FLASHING CONTROL SIGNAL 3.50%
LANE USE MARKING        2.62%
NO CONTROLS             1.42%
NO PASSING              5.08%
OTHER                   2.38%
OTHER RAILROAD CROSSING 2.55%
OTHER REG. SIGN         2.28%
OTHER WARNING SIGN      2.31%
PEDESTRIAN CROSSING SIGN 9.25%
POLICE/FLAGMAN          2.82%
RAILROAD CROSSING GATE  2.51%
RR CROSSING SIGN        0.98%
SCHOOL ZONE             2.95%
STOP SIGN/FLASHER       2.34%
TRAFFIC SIGNAL          2.30%
UNKNOWN                 1.15%
YIELD                   3.30%
```

7. Pig Analysis

Apache Pig was used to analyze the traffic crash dataset using operations such as filtering, grouping, counting, summation, sorting and ranking.

7.1 Top 10 Accident-Prone Streets

Top 10 Accident-Prone Streets: Returns the 10 streets with the highest number of traffic crashes.

Script

```
infile = LOAD 'C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv'
USING PigStorage(',')
AS
(crash_record_id:chararray,crash_date:chararray,posted_speed_limit:chararray,traffic_control_device:chararray, device_condition:chararray,
weather_condition:chararray,lighting_condition:chararray,first_crash_type:chararray,trafficway_type:chararray,
roadway_surface_cond:chararray,road_defect:chararray,crash_type:chararray,damage:chararray,prim_contributory_cause:chararray,
street_direction:chararray,street_name:chararray,num_units:chararray,most_severe_injury:chararray,injuries_total:chararray,
injuries_fatal:chararray,injuries_incapacitating:chararray,crash_hour:chararray,crash_day_of_week:chararray,crash_month:chararray);
files = FILTER infile BY
    crash_record_id != 'CRASH_RECORD_ID'
    AND street_name IS NOT NULL
    AND street_name != '';
grouped_street = GROUP files BY street_name;
street_count = FOREACH grouped_street
GENERATE
    group AS street_name,
    COUNT(files) AS total_crashes;
order_street = ORDER street_count
    BY total_crashes DESC;
top10_street = LIMIT order_street 10;
STORE top10_street
INTO 'C:/BDAL/BDAPProject/PIGFolder/Top10AccidentStreets'
USING PigStorage('|');
```

Execution

```
C:\BDAL\BDAPProject>cd C:\BDAL\BDAPProject\PIGFolder
C:\BDAL\BDAPProject\PIGFolder>pig -x local Top10AccidentStreets.pig
2026-09-10 15:24:53,334 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-10 15:24:53,334 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-10 15:24:53,757 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-10 15:24:53,757 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789032293753.log
2026-09-10 15:24:53,792 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated. Instead, use mapreduce.job.user.name
2026-09-10 15:24:53,963 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Biswas/.pigbootstrap not found
2026-09-10 15:24:54,046 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-10 15:24:54,088 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: file:///
2026-09-10 15:24:54,088 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-Top10AccidentStreets.pig-ee3cealc-8cfa-451c-b142-72e30eb5b0e8
2026-09-10 15:24:54,930 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-10 15:24:54,930 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.text
outputformat.separator
2026-09-10 15:24:54,955 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP,BY,ORDER_BY,FILTER,LIMIT
2026-09-10 15:24:55,038 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator
, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, Pr
edicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-10 15:24:55,124 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 4
89580128, usageThreshold = 489580128
2026-09-10 15:24:55,218 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-10 15:24:55,277 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-10 15:24:55,307 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce r
educe scope=102
2026-09-10 15:24:55,327 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-10 15:24:55,327 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-10 15:24:55,632 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-10 15:24:55,640 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.re
duce.markreset.buffer.percent
```

```

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  Me
ure  Outputs
job_local1135289706_0002  1  1  1  n/a  n/a  n/a  n/a  n/a  n/a  order_street  SAMPLER
job_local281286527_0001  11  1  1  n/a  n/a  n/a  n/a  n/a  n/a  files,grouped_street,infiles,stre

job_local80988267_0003  1  1  1  n/a  n/a  n/a  n/a  n/a  n/a  order_street  ORDER_BY,COMBINER
job_local826914119_0004  1  1  1  n/a  n/a  n/a  n/a  n/a  n/a  order_street  C:/BDAL/BD

reets.txt,

Input(s):
Successfully read 936812 records from: "C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv"

Output(s):
Successfully stored 10 records in: "C:/BDAL/BDAPProject/PIGFolder/Top10AccidentStreets.txt"

Counters:
Total records written : 10
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local281286527_0001 -> job_local1135289706_0002,
job_local1135289706_0002 -> job_local80988267_0003,
job_local80988267_0003 -> job_local826914119_0004,
job_local826914119_0004

2026-09-10 15:25:09,349 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:25:09,362 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:25:09,365 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:25:09,384 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:25:09,389 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:25:09,393 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:25:09,398 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!

```

Output

```

C:\BDAL\BDAPProject\PIGFolder>dir Top10AccidentStreets\*
Volume in drive C has no label.
Volume Serial Number is DA25-CDAA0

Directory of C:\BDAL\BDAPProject\PIGFolder\Top10AccidentStreets

09/10/2026  03:25 PM  <DIR>          .
09/10/2026  04:22 PM  <DIR>          ..
09/10/2026  03:25 PM                12  .part-r-00000.crc
09/10/2026  03:25 PM                8  .._SUCCESS.crc
09/10/2026  03:25 PM             169  part-r-00000
09/10/2026  03:25 PM                0  _SUCCESS
               4 File(s)              189 bytes
               2 Dir(s)  129,430,892,544 bytes free

C:\BDAL\BDAPProject\PIGFolder>type Top10AccidentStreets\part-r-00000
WESTERN AVE|25645
PULASKI RD|22603
CICERO AVE|21024
ASHLAND AVE|20373
HALSTED ST|18091
KEDZIE AVE|16388
MICHIGAN AVE|11996
NORTH AVE|10793
STATE ST|10597
CLARK ST|10035

```

7.2 Top 10 Primary Causes of Crashes

Top 10 Primary Causes of Crashes: Returns the 10 most common primary contributing causes of traffic crashes.

Script

```
infile = LOAD 'C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv'
USING PigStorage(',')
AS
(crash_record_id:chararray,crash_date:chararray,posted_speed_limit:chararray,traffic_control_device:chararray, device_condition:chararray,
weather_condition:chararray,lighting_condition:chararray,first_crash_type:chararray,trafficway_type:chararray,
roadway_surface_cond:chararray,road_defect:chararray,crash_type:chararray,damage:chararray,prim_contributory_cause:chararray,
street_direction:chararray,street_name:chararray,num_units:chararray,most_severe_injury:chararray,injuries_total:chararray,
injuries_fatal:chararray,injuries_incapacitating:chararray,crash_hour:chararray,crash_day_of_week:chararray,crash_month:chararray);
files = FILTER infile BY
    crash_record_id != 'CRASH_RECORD_ID'
    AND prim_contributory_cause IS NOT NULL
    AND prim_contributory_cause != '';
grouped_cause = GROUP files BY prim_contributory_cause;
cause_count = FOREACH grouped_cause
    GENERATE
        group AS primary_cause,
        COUNT(files) AS total_crashes;
order_cause = ORDER cause_count BY total_crashes DESC;
top10_cause = LIMIT order_cause 10;
STORE top10_cause
INTO 'C:/BDAL/BDAPProject/PIGFolder/Top10PrimaryCauses'
USING PigStorage('|');
```

Execution

```
Administrator: Command Prompt

C:\BDAL\BDAPProject\PIGFolder>pig -x local Top10PrimaryCauses.pig
2026-09-10 15:39:07,965 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-10 15:39:07,965 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-10 15:39:08,406 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-10 15:39:08,407 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789033148402.log
2026-09-10 15:39:08,433 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated. Instead, use mapreduce.job.user.name
2026-09-10 15:39:08,604 [main] INFO org.apache.pig.impl.util.Utils - Default bootup file C:\Users\Biswas/.pigbootup not found
2026-09-10 15:39:08,677 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-10 15:39:08,680 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: file:///
2026-09-10 15:39:08,713 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-Top10PrimaryCauses.pig-1972792b-d7cf-44bf-8a04-13811be7057e
2026-09-10 15:39:08,713 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-10 15:39:09,457 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.outp
utputformat.separator
2026-09-10 15:39:09,484 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER,LIMIT
2026-09-10 15:39:09,572 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCa
l, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptim
edicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-10 15:39:09,663 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThres
89580128, usageThreshold = 489580128
2026-09-10 15:39:09,757 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-10 15:39:09,816 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-10 15:39:09,833 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for Map
ode scope-102
2026-09-10 15:39:09,849 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-10 15:39:09,849 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-10 15:39:10,181 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-10 15:39:10,198 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapre
uce.markreset.buffer.percent
2026-09-10 15:39:10,198 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is
, set to default 0.3
2026-09-10 15:39:10,201 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutp
.compress
2026-09-10 15:39:10,204 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of requir
ers.
2026-09-10 15:39:10,206 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backe
p.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-10 15:39:10,214 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers
allInputFileSize=345214189
2026-09-10 15:39:10,214 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
```

```

Administrator: Command Prompt
Success!
Job Stats (time in seconds):
JobId  Maps  Reduces MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  MedianReducetime  Alias  Feat
ure  Outputs
job_local1548345365_0004  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  order_cause  C:/BDAL/BDAPProject/PIGFolder/Top10Pr
maryCauses.txt,
job_local1635532600_0002  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  order_cause  SAMPLER
job_local1764445225_0001  11  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  cause_count,files,grouped_cause,infiles GROUP_BY,COMBINER
job_local191904975_0003  1  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  order_cause  ORDER_BY,COMBINER

Input(s):
Successfully read 936812 records from: "C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv"

Output(s):
Successfully stored 10 records in: "C:/BDAL/BDAPProject/PIGFolder/Top10PrimaryCauses.txt"

Counters:
Total records written : 10
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1764445225_0001 -> job_local1635532600_0002,
job_local1635532600_0002 -> job_local191904975_0003,
job_local191904975_0003 -> job_local1548345365_0004,
job_local1548345365_0004

```

Output

```

C:\BDAL\BDAPProject\PIGFolder>dir Top10PrimaryCauses\*
Volume in drive C has no label.
Volume Serial Number is DA25-CDA0

Directory of C:\BDAL\BDAPProject\PIGFolder\Top10PrimaryCauses

09/10/2026  03:39 PM  <DIR>          .
09/10/2026  04:22 PM  <DIR>          ..
09/10/2026  03:39 PM                12 .part-r-00000.crc
09/10/2026  03:39 PM                8 ._SUCCESS.crc
09/10/2026  03:39 PM            316 part-r-00000
09/10/2026  03:39 PM                0 _SUCCESS
               4 File(s)            336 bytes
               2 Dir(s) 129,429,925,888 bytes free

C:\BDAL\BDAPProject\PIGFolder>type Top10PrimaryCauses\part-r-00000
UNABLE TO DETERMINE|367138
FAILING TO YIELD RIGHT-OF-WAY|103739
FOLLOWING TOO CLOSELY|90033
NOT APPLICABLE|49213
IMPROPER OVERTAKING/PASSING|46802
FAILING TO REDUCE SPEED TO AVOID CRASH|39308
IMPROPER BACKING|35900
IMPROPER LANE USAGE|33252
DRIVING SKILLS/KNOWLEDGE/EXPERIENCE|32027
IMPROPER TURNING/NO SIGNAL|31418

```

7.3 Top 10 Dangerous Hours by Total Injuries

Top 10 Dangerous Hours by Total Injuries: Returns the 10 hours of the day with the highest total number of injuries.

Script

```
infile = LOAD 'C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv'
USING PigStorage(',')
AS
(crash_record_id:chararray,crash_date:chararray,posted_speed_limit:chararray,traffic_control_device:chararray, device_condition:chararray,
weather_condition:chararray,lighting_condition:chararray,first_crash_type:chararray,trafficway_type:chararray,
roadway_surface_cond:chararray,road_defect:chararray,crash_type:chararray,damage:chararray,prim_contributory_cause:chararray,
street_direction:chararray,street_name:chararray,num_units:chararray,most_severe_injury:chararray,injuries_total:chararray,
injuries_fatal:chararray,injuries_incapacitating:chararray,crash_hour:chararray,crash_day_of_week:chararray,crash_month:chararray);
files = FILTER infile BY
    crash_record_id != 'CRASH_RECORD_ID'
    AND crash_hour IS NOT NULL
    AND crash_hour != ''
    AND injuries_total IS NOT NULL
    AND injuries_total != '';
prepared = FOREACH files GENERATE
    (int)crash_hour AS crash_hour,
    (double)injuries_total AS injuries_total;
grouped_hour = GROUP prepared BY crash_hour;
hour_injuries = FOREACH grouped_hour
    GENERATE
        group AS crash_hour,
        SUM(prepared.injuries_total) AS total_injuries;
order_hour = ORDER hour_injuries BY total_injuries DESC;
top10_hour = LIMIT order_hour 10;
STORE top10_hour
INTO 'C:/BDAL/BDAPProject/PIGFolder/Top10DangerousHours'
USING PigStorage(',');
```

Execution

```
Administrator: Command Prompt

Success!

Job Stats (time in seconds):
JobId  Maps  Reduces  MaxMapTime  MinMapTime  AvgMapTime  MedianMapTime  MaxReduceTime  MinReduceTime  AvgReduceTime  Me
ure  Outputs
job_local1067886893_0001  11  1  n/a  n/a  n/a  n/a  n/a  n/a  files,grouped_hour,hour_in
P_BY,COMBINER
job_local1393830340_0003  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ORDER_BY,COMBINER
job_local1552683396_0002  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ORDER_BY,SAMPLER
job_local1653020975_0004  1  n/a  n/a  n/a  n/a  n/a  n/a  n/a  ORDER_BY,SAMPLER
C:/BDAL/BD
ours.txt,

Input(s):
Successfully read 936812 records from: "C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv"

Output(s):
Successfully stored 10 records in: "C:/BDAL/BDAPProject/PIGFolder/Top10DangerousHours.txt"

Counters:
Total records written : 10
Total bytes written : 0
Spillable Memory Manager spill count : 0
Total bags proactively spilled: 0
Total records proactively spilled: 0

Job DAG:
job_local1067886893_0001 -> job_local1552683396_0002,
job_local1552683396_0002 -> job_local1393830340_0003,
job_local1393830340_0003 -> job_local1653020975_0004,
job_local1653020975_0004

2026-09-10 15:44:58,652 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:44:58,664 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:44:58,667 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:44:58,680 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:44:58,686 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:44:58,689 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
2026-09-10 15:44:58,697 [main] WARN org.apache.hadoop.metrics2.impl.MetricsSystemImpl - JobTracker metrics system already initialized!
```

Output

```
C:\BDAL\BDAPProject\PIGFolder>dir Top10DangerousHours\*
Volume in drive C has no label.
Volume Serial Number is DA25-CDA0

Directory of C:\BDAL\BDAPProject\PIGFolder\Top10DangerousHours

09/10/2026  03:44 PM    <DIR>          .
09/10/2026  04:22 PM    <DIR>          ..
09/10/2026  03:44 PM                12 .part-r-00000.crc
09/10/2026  03:44 PM                8  ._SUCCESS.crc
09/10/2026  03:44 PM               105 part-r-00000
09/10/2026  03:44 PM                0 _SUCCESS
                4 File(s)              125 bytes
                2 Dir(s)  129,429,880,832 bytes free

C:\BDAL\BDAPProject\PIGFolder>type Top10DangerousHours\part-r-00000
17|13052.0
16|13011.0
15|12878.0
18|11242.0
14|11151.0
13|10061.0
12|9168.0
19|9068.0
8|8613.0
11|8262.0
```

7.4 Monthly Crash Trend

Monthly Crash Trend: Returns the total number of traffic crashes for each month.

Script

```
infile = LOAD 'C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv'
USING PigStorage(',')
AS
(crash_record_id:chararray,crash_date:chararray,posted_speed_limit:chararray,traffic_control_device:chararray, device_condition:chararray,
weather_condition:chararray,lighting_condition:chararray,first_crash_type:chararray,trafficway_type:chararray,
roadway_surface_cond:chararray,road_defect:chararray,crash_type:chararray,damage:chararray,prim_contributory_cause:chararray,
street_direction:chararray,street_name:chararray,num_units:chararray,most_severe_injury:chararray,injuries_total:chararray,
injuries_fatal:chararray,injuries_incapacitating:chararray,crash_hour:chararray,crash_day_of_week:chararray,crash_month:chararray);
files = FILTER infile BY
    crash_record_id != 'CRASH_RECORD_ID'
    AND crash_month IS NOT NULL
    AND crash_month != '';
prepared = FOREACH files GENERATE
    (int)crash_month AS crash_month;
grouped_month = GROUP prepared BY crash_month;
month_count = FOREACH grouped_month
    GENERATE
        group AS crash_month,
        COUNT(prepared) AS total_crashes;
order_month = ORDER month_count BY crash_month ASC;
STORE order_month
INTO 'C:/BDAL/BDAPProject/PIGFolder/MonthlyCrashTrend'
USING PigStorage('|');
```

Execution

```
Administrator: Command Prompt
C:\BDAL\BDAPProject\PIGFolder>pig -x local MonthlyCrashTrend.pig
2026-09-10 15:49:45,123 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-10 15:49:45,123 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-10 15:49:45,543 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-10 15:49:45,543 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789033785539.log
2026-09-10 15:49:45,572 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated. Instead, use mapreduce.job.user.name
2026-09-10 15:49:45,767 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Biswas/.pigbootstrap not found
2026-09-10 15:49:45,837 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-10 15:49:45,840 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: file:///
2026-09-10 15:49:45,874 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-MonthlyCrashTrend.pig-8d5fff56-9705-4fe8-b5f5-5b2b2a2a6e5d
2026-09-10 15:49:45,875 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-10 15:49:46,659 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-10 15:49:46,681 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER
2026-09-10 15:49:46,759 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSet, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-10 15:49:46,794 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned for infiles: $1, $2, $3, $4, $5, $6, $7, $8, $9, $10, $11, $12, $13, $14, $15, $16, $17, $18, $19, $20, $21, $22
2026-09-10 15:49:46,842 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-10 15:49:46,921 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-10 15:49:46,989 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-10 15:49:47,014 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope-40
2026-09-10 15:49:47,032 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-09-10 15:49:47,032 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-10 15:49:47,365 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-10 15:49:47,380 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-10 15:49:47,380 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-10 15:49:47,989 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-10 15:49:47,382 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-10 15:49:47,382 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-10 15:49:47,398 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=100000000 maxReducers=999 tot
```

Success!

Job Stats (time in seconds):

JobId	Maps	Reduces	MaxMapTime	MinMapTime	AvgMapTime	MedianMapTime	MaxReduceTime	MinReduceTime
job_local1548054697_0003	1	1	n/a	n/a	n/a	n/a	n/a	n/a
job_local605660327_0001	11	1	n/a	n/a	n/a	n/a	n/a	n/a
job_local667134317_0002	1	1	n/a	n/a	n/a	n/a	n/a	n/a

Input(s):

Successfully read 936812 records from: "C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv"

Output(s):

Successfully stored 12 records in: "C:/BDAL/BDAPProject/PIGFolder/MonthlyCrashTrend.txt"

Counters:

Total records written : 12

Total bytes written : 0

Spillable Memory Manager spill count : 0

Total bags proactively spilled: 0

Total records proactively spilled: 0

Job DAG:

```
job_local605660327_0001 -> job_local667134317_0002,
job_local667134317_0002 -> job_local1548054697_0003,
job_local1548054697_0003
```

Output

```
C:\BDAL\BDAPProject\PIGFolder>dir MonthlyCrashTrend\*
Volume in drive C has no label.
Volume Serial Number is DA25-CDA0

Directory of C:\BDAL\BDAPProject\PIGFolder\MonthlyCrashTrend

09/10/2026  03:49 PM    <DIR>          .
09/10/2026  04:22 PM    <DIR>          ..
09/10/2026  03:49 PM                12 .part-r-00000.crc
09/10/2026  03:49 PM                 8 ._SUCCESS.crc
09/10/2026  03:49 PM                99 part-r-00000
09/10/2026  03:49 PM                 0 _SUCCESS
               4 File(s)              119 bytes
               2 Dir(s) 129,429,868,544 bytes free

C:\BDAL\BDAPProject\PIGFolder>type MonthlyCrashTrend\part-r-00000
1|74281
2|73047
3|76539
4|71837
5|77269
6|77697
7|78571
8|80826
9|82232
10|86692
11|78203
12|79617
```

7.5 Day-of-Week Crash Analysis

Day-of-Week Crash Analysis: Returns the total number of crashes for each day of the week.

Script

```
infiles = LOAD 'C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv'
USING PigStorage(',')
AS
(crash_record_id:chararray,crash_date:chararray,posted_speed_limit:chararray,traffic_control_device:chararray, device_condition:chararray,
weather_condition:chararray,lighting_condition:chararray,first_crash_type:chararray,trafficway_type:chararray,
roadway_surface_cond:chararray,road_defect:chararray,crash_type:chararray,damage:chararray,prim_contributory_cause:chararray,
street_direction:chararray,street_name:chararray,num_units:chararray,most_severe_injury:chararray,injuries_total:chararray,
injuries_fatal:chararray,injuries_incapacitating:chararray,crash_hour:chararray,crash_day_of_week:chararray,crash_month:chararray);
files = FILTER infiles BY
    crash_record_id != 'CRASH_RECORD_ID'
    AND crash_day_of_week IS NOT NULL
    AND crash_day_of_week != '';
prepared = FOREACH files GENERATE
    (int)crash_day_of_week AS crash_day_of_week;
grouped_day = GROUP prepared BY crash_day_of_week;
day_count = FOREACH grouped_day
    GENERATE
        group AS crash_day_of_week,
        COUNT(prepared) AS total_crashes;
order_day = ORDER day_count BY total_crashes DESC;
STORE order_day
INTO 'C:/BDAL/BDAPProject/PIGFolder/DayOfWeekCrashAnalysis'
USING PigStorage('|');
```

Execution

```
C:\BDAL\BDAPProject\PIGFolder>pig -x local DayOfWeekCrashAnalysis.pig
2026-09-10 15:55:25,569 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-10 15:55:25,571 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-10 15:55:25,854 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-10 15:55:25,854 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789034125851.log
2026-09-10 15:55:25,871 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated. Instead, use mapreduce.job.user.name
2026-09-10 15:55:25,982 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Biswas/.pigbootstrap not found
2026-09-10 15:55:26,043 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-10 15:55:26,045 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: file:///
2026-09-10 15:55:26,071 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-DayOfWeekCrashAnalysis.pig-8ed92fa5-2dc3-4138-81c7-dd8114fcc5ce
2026-09-10 15:55:26,071 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-10 15:55:26,562 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-10 15:55:26,574 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP_BY,ORDER_BY,FILTER
2026-09-10 15:55:26,626 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-10 15:55:26,645 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned for infiles: $1, $2, $3, $4, $5, $6, $7, $8, $9, $10, $11, $12, $13, $14, $15, $16, $17, $18, $19, $20, $21, $23
2026-09-10 15:55:26,681 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-10 15:55:26,735 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-10 15:55:26,770 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-10 15:55:26,792 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope-40
2026-09-10 15:55:26,804 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-09-10 15:55:26,804 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-10 15:55:27,019 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-10 15:55:27,027 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-10 15:55:27,027 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-10 15:55:27,031 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-10 15:55:27,034 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers
2026-09-10 15:55:27,035 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-10 15:55:27,043 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 tot
```

CA Administrator: Command Prompt

Success!

Job Stats (time in seconds):

JobId	Maps	Reduces	MaxMapTime	MinMapTime	AvgMapTime	MedianMapTime	MaxReduceTime	Min
job_local1355628479_0001	11	1	n/a	n/a	n/a	n/a	n/a	n/a
job_local26340013_0003	1	1	n/a	n/a	n/a	n/a	n/a	ord
job_local271711779_0002	1	1	n/a	n/a	n/a	n/a	n/a	ord

Input(s):

Successfully read 936812 records from: "C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv"

Output(s):

Successfully stored 7 records in: "C:/BDAL/BDAPProject/PIGFolder/DayOfWeekCrashAnalysis.txt"

Counters:

Total records written : 7

Total bytes written : 0

Spillable Memory Manager spill count : 0

Total bags proactively spilled: 0

Total records proactively spilled: 0

Job DAG:

```
job_local1355628479_0001 -> job_local271711779_0002,
job_local271711779_0002 -> job_local26340013_0003,
job_local26340013_0003
```

Output

```
C:\BDAL\BDAPProject\PIGFolder>dir DayOfWeekCrashAnalysis\*
Volume in drive C has no label.
Volume Serial Number is DA25-CDA0

Directory of C:\BDAL\BDAPProject\PIGFolder\DayOfWeekCrashAnalysis

09/10/2026  03:55 PM    <DIR>          .
09/10/2026  04:22 PM    <DIR>          ..
09/10/2026  03:55 PM                12 .part-r-00000.crc
09/10/2026  03:55 PM                8 ._SUCCESS.crc
09/10/2026  03:55 PM               63 part-r-00000
09/10/2026  03:55 PM                0 _SUCCESS
                4 File(s)                83 bytes
                2 Dir(s)  129,430,044,672 bytes free

C:\BDAL\BDAPProject\PIGFolder>type DayOfWeekCrashAnalysis\part-r-00000
6|152231
7|138210
5|135323
3|133327
4|132819
2|128340
1|116561
```

7.6 Top Crash Types

Top Crash Types: Returns the most frequently recorded types of traffic crashes.

Script

```
infile = LOAD 'C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv'
USING PigStorage(',')
AS
(crash_record_id:chararray,crash_date:chararray,posted_speed_limit:chararray,traffic_control_device:chararray, device_condition:chararray,
weather_condition:chararray,lighting_condition:chararray,first_crash_type:chararray,trafficway_type:chararray,
roadway_surface_cond:chararray,road_defect:chararray,crash_type:chararray,damage:chararray,prim_contributory_cause:chararray,
street_direction:chararray,street_name:chararray,num_units:chararray,most_severe_injury:chararray,injuries_total:chararray,
injuries_fatal:chararray,injuries_incapacitating:chararray,crash_hour:chararray,crash_day_of_week:chararray,crash_month:chararray);
files = FILTER infile BY
    crash_record_id != 'CRASH_RECORD_ID'
    AND first_crash_type IS NOT NULL
    AND first_crash_type != '';
grouped_type = GROUP files BY first_crash_type;
type_count = FOREACH grouped_type
    GENERATE
        group AS crash_type,
        COUNT(files) AS total_crashes;
order_type = ORDER type_count BY total_crashes DESC;
top10_type = LIMIT order_type 10;
STORE top10_type
INTO 'C:/BDAL/BDAPProject/PIGFolder/TopCrashTypes'
USING PigStorage('|');
```

Execution

```
Select Administrator: Command Prompt
C:\BDAL\BDAPProject\PIGFolder>pig -x local TopCrashTypes.pig
2026-09-10 16:00:46,379 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-10 16:00:46,379 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-10 16:00:46,656 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-10 16:00:46,656 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789034446653.log
2026-09-10 16:00:46,677 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated. Instead, use mapreduce.job.user.name
2026-09-10 16:00:46,794 [main] INFO org.apache.pig.impl.util.Util - Default bootstrap file C:\User\Biswas/.pigbootstrap not found
2026-09-10 16:00:46,841 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-10 16:00:46,841 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: file:///
2026-09-10 16:00:46,873 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-TopCrashTypes.pig-bb0c2379-3536-4317-a66b-9beb96e7a0e1
2026-09-10 16:00:46,873 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-10 16:00:47,367 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-10 16:00:47,382 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP BY, ORDER BY, FILTER, LIMIT
2026-09-10 16:00:47,430 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSet, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-10 16:00:47,478 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-10 16:00:47,542 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-10 16:00:47,574 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombineOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-10 16:00:47,590 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce reduce scope=102
2026-09-10 16:00:47,622 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-10 16:00:47,622 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-10 16:00:47,828 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-10 16:00:47,828 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-10 16:00:47,828 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-10 16:00:47,828 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-10 16:00:47,844 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers
2026-09-10 16:00:47,844 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-10 16:00:47,844 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=345214189
2026-09-10 16:00:47,844 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
```

Select Administrator: Command Prompt

Success!

Job Stats (time in seconds):

JobId	Maps	Reduces	MaxMapTime	MinMapTime	AvgMapTime	MedianMapTime	MaxReduceTime
job_local1052041930_0004	1	1	n/a	n/a	n/a	n/a	n/a
job_local1187387753_0002	1	1	n/a	n/a	n/a	n/a	n/a
job_local11786944715_0001	11	1	n/a	n/a	n/a	n/a	n/a
job_local640998836_0003	1	1	n/a	n/a	n/a	n/a	n/a

Input(s):

Successfully read 936812 records from: "C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv"

Output(s):

Successfully stored 10 records in: "C:/BDAL/BDAPProject/PIGFolder/TopCrashTypes.txt"

Counters:

Total records written : 10

Total bytes written : 0

Spillable Memory Manager spill count : 0

Total bags proactively spilled: 0

Total records proactively spilled: 0

Job DAG:

```
job_local11786944715_0001 -> job_local1187387753_0002,
job_local1187387753_0002 -> job_local640998836_0003,
job_local640998836_0003 -> job_local1052041930_0004,
job_local1052041930_0004
```

Output

```
C:\BDAL\BDAPProject\PIGFolder>dir TopCrashTypes\*
Volume in drive C has no label.
Volume Serial Number is DA25-CDA0

Directory of C:\BDAL\BDAPProject\PIGFolder\TopCrashTypes

09/10/2026  04:00 PM    <DIR>          .
09/10/2026  04:22 PM    <DIR>          ..
09/10/2026  04:00 PM                12 .part-r-00000.crc
09/10/2026  04:00 PM                8 ._SUCCESS.crc
09/10/2026  04:00 PM               213 part-r-00000
09/10/2026  04:00 PM                0 _SUCCESS
               4 File(s)                233 bytes
               2 Dir(s) 129,429,544,960 bytes free

C:\BDAL\BDAPProject\PIGFolder>type TopCrashTypes\part-r-00000
PARKED MOTOR VEHICLE|216715
REAR END|206197
SIDESWIPE SAME DIRECTION|144162
TURNING|135089
ANGLE|102107
FIXED OBJECT|43465
PEDESTRIAN|22313
PEDALCYCLIST|14700
SIDESWIPE OPPOSITE DIRECTION|12967
REAR TO FRONT|9841
```

7.7 Fatality Ranking by Crash Type and Road Condition

Fatality Ranking by Crash Type and Road Condition: Returns the crash-type and road-surface combinations with the highest total number of fatalities.

Script

```
infile = LOAD 'C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv'
USING PigStorage(',')
AS
(crash_record_id:chararray,crash_date:chararray,posted_speed_limit:chararray,traffic_control_device:chararray, device_condition:chararray,
weather_condition:chararray,lighting_condition:chararray,first_crash_type:chararray,trafficway_type:chararray,
roadway_surface_cond:chararray,road_defect:chararray,crash_type:chararray,damage:chararray,prim_contributory_cause:chararray,
street_direction:chararray,street_name:chararray,num_units:chararray,most_severe_injury:chararray,injuries_total:chararray,
injuries_fatal:chararray,injuries_incapacitating:chararray,crash_hour:chararray,crash_day_of_week:chararray,crash_month:chararray);
files = FILTER infile BY
    crash_record_id != 'CRASH_RECORD_ID'
    AND first_crash_type IS NOT NULL
    AND first_crash_type != ''
    AND roadway_surface_cond IS NOT NULL
    AND roadway_surface_cond != ''
    AND injuries_fatal IS NOT NULL
    AND injuries_fatal != '';
prepared = FOREACH files GENERATE
    first_crash_type,
    roadway_surface_cond,
    (double)injuries_fatal AS fatalities;
grouped_data = GROUP prepared BY
    (first_crash_type, roadway_surface_cond);
fatality_count = FOREACH grouped_data
    GENERATE
    FLATTEN(group) AS
    (first_crash_type, roadway_surface_cond),
    SUM(prepared.fatalities) AS total_fatalities;
positive_fatalities = FILTER fatality_count BY
    total_fatalities > 0;
order_fatality = ORDER positive_fatalities
    BY total_fatalities DESC;
top10_fatality = LIMIT order_fatality 10;
STORE top10_fatality
INTO 'C:/BDAL/BDAPProject/PIGFolder/FatalityRanking'
USING PigStorage('|');
```

Execution

```
Administrator: Command Prompt
C:\BDAL\BDAPProject>pig -x local FatalityRanking.pig
2026-09-10 16:14:29,195 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-10 16:14:29,195 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-10 16:14:29,479 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-10 16:14:29,479 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789035269476.log
2026-09-10 16:14:29,499 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated. Instead, use mapreduce.job.user.name
2026-09-10 16:14:29,615 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Biswas/.pigbootstrap not found
2026-09-10 16:14:29,667 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-10 16:14:29,669 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: file:///
2026-09-10 16:14:29,698 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-FatalityRanking.pig-7100f1b9-4881-46ed-bbbb-26db3377925f
2026-09-10 16:14:29,698 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-10 16:14:30,238 [main] WARN org.apache.pig.newplan.BaseOperatorPlan - Encountered Warning IMPLICIT_CAST_TO_DOUBLE 1 time(s).
2026-09-10 16:14:30,244 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-10 16:14:30,260 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig Features used in the script: GROUP_BY,ORDER_BY,FILTER,LIMIT
2026-09-10 16:14:30,313 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-10 16:14:30,346 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-10 16:14:30,384 [main] INFO org.apache.pig.newplan.logical.rules.ColumnPruneVisitor - Columns pruned for infiles: $1, $2, $3, $4, $5, $6, $8, $10, $11, $12, $13, $14, $15, $16, $17, $18, $20, $21, $22, $23
2026-09-10 16:14:30,438 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-10 16:14:30,483 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-10 16:14:30,500 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce node scope-74
2026-09-10 16:14:30,510 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 4
2026-09-10 16:14:30,510 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 4
2026-09-10 16:14:30,715 [main] INFO org.apache.pig.tools.pigstats.MapReduce.MRScriptState - Pig script settings are added to the job
2026-09-10 16:14:30,731 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-10 16:14:30,731 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-10 16:14:30,731 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-10 16:14:30,731 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-10 16:14:30,731 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
```

Success!

Job Stats (time in seconds):

JobId	Maps	Reduces	MaxMapTime	MinMapTime	AvgMapTime	MedianMapTime	MaxReduceTime
job_local1307034661_0004	1	1	n/a	n/a	n/a	n/a	n/a
job_local1348348206_0001	11	1	n/a	n/a	n/a	n/a	n/a
job_local1886177834_0003	1	1	n/a	n/a	n/a	n/a	n/a
job_local1987705024_0002	1	1	n/a	n/a	n/a	n/a	n/a

Input(s):

Successfully read 936812 records from: "C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv"

Output(s):

Successfully stored 10 records in: "C:/BDAL/BDAPProject/PIGFolder/FatalityRanking"

Counters:

Total records written : 10

Total bytes written : 0

Spillable Memory Manager spill count : 0

Total bags proactively spilled: 0

Total records proactively spilled: 0

Job DAG:

```
job_local1348348206_0001 -> job_local1987705024_0002,
job_local1987705024_0002 -> job_local1886177834_0003,
job_local1886177834_0003 -> job_local1307034661_0004,
job_local1307034661_0004
```

Output

```
C:\BDAL\BDAPProject\PIGFolder>dir FatalityRanking\*
Volume in drive C has no label.
Volume Serial Number is DA25-CDA0

Directory of C:\BDAL\BDAPProject\PIGFolder\FatalityRanking

09/10/2026  04:14 PM    <DIR>          .
09/10/2026  04:14 PM    <DIR>          ..
09/10/2026  04:14 PM                12 .part-r-00000.crc
09/10/2026  04:14 PM                 8 ._SUCCESS.crc
09/10/2026  04:14 PM            206 part-r-00000
09/10/2026  04:14 PM                 0 _SUCCESS
               4 File(s)            226 bytes
               2 Dir(s) 129,436,438,528 bytes free

C:\BDAL\BDAPProject\PIGFolder>type FatalityRanking\part-r-00000
PEDESTRIAN|DRY|211.0
FIXED OBJECT|DRY|198.0
ANGLE|DRY|112.0
TURNING|DRY|83.0
PARKED MOTOR VEHICLE|DRY|78.0
FIXED OBJECT|WET|45.0
HEAD ON|DRY|44.0
PEDESTRIAN|WET|39.0
REAR END|DRY|37.0
PEDALCYCLIST|DRY|31.0
```

7.8 Most Severe Injury Distribution

Most Severe Injury Distribution: Returns the total number of crashes for each most-severe-injury category.

Script

```
infile = LOAD 'C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv'
USING PigStorage(',')
AS
(crash_record_id:chararray,crash_date:chararray,posted_speed_limit:chararray,traffic_control_device:chararray, device_condition:chararray,
weather_condition:chararray,lighting_condition:chararray,first_crash_type:chararray,trafficway_type:chararray,
roadway_surface_cond:chararray,road_defect:chararray,crash_type:chararray,damage:chararray,prim_contributory_cause:chararray,
street_direction:chararray,street_name:chararray,num_units:chararray,most_severe_injury:chararray,injuries_total:chararray,
injuries_fatal:chararray,injuries_incapacitating:chararray,crash_hour:chararray,crash_day_of_week:chararray,crash_month:chararray);
files = FILTER infile BY
    crash_record_id != 'CRASH_RECORD_ID'
    AND most_severe_injury IS NOT NULL
    AND most_severe_injury != '';

grouped_injury = GROUP files BY most_severe_injury;

injury_count = FOREACH grouped_injury
    GENERATE
        group AS injury_type,
        COUNT(files) AS total_crashes;

order_injury = ORDER injury_count
    BY total_crashes DESC;

STORE order_injury
    INTO 'C:/BDAL/BDAPProject/PIGFolder/SevereInjuryDistribution'
    USING PigStorage('|');
```

Execution

```
Administrator: Command Prompt
C:\BDAL\BDAPProject\PIGFolder>pig -x local SevereInjuryDistribution.pig
2026-09-10 16:22:12,481 INFO pig.ExecTypeProvider: Trying ExecType : LOCAL
2026-09-10 16:22:12,482 INFO pig.ExecTypeProvider: Picked LOCAL as the ExecType
2026-09-10 16:22:12,696 [main] INFO org.apache.pig.Main - Apache Pig version 0.18.0 (r1926484) compiled Jun 16 2025, 10:38:21
2026-09-10 16:22:12,696 [main] INFO org.apache.pig.Main - Logging error messages to: C:\hadoop\logs\pig_1789035732693.log
2026-09-10 16:22:12,720 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - user.name is deprecated. Instead, use mapreduce.job.user.name
2026-09-10 16:22:12,843 [main] INFO org.apache.pig.impl.util.Utils - Default bootstrap file C:\Users\Biswas/.pigbootstrap not found
2026-09-10 16:22:12,893 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.tracker is deprecated. Instead, use mapreduce.jobtracker.address
2026-09-10 16:22:12,895 [main] INFO org.apache.pig.backend.hadoop.executionengine.HExecutionEngine - Connecting to hadoop file system at: file:///
2026-09-10 16:22:12,920 [main] INFO org.apache.pig.PigServer - Pig Script ID for the session: PIG-SevereInjuryDistribution.pig-80e02bfa-ec76-4641-9940-b6de939198b9
2026-09-10 16:22:12,920 [main] WARN org.apache.pig.PigServer - ATS is disabled since yarn.timeline-service.enabled set to false
2026-09-10 16:22:13,395 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.textoutputformat.separator is deprecated. Instead, use mapreduce.output.textoutputformat.separator
2026-09-10 16:22:13,409 [main] INFO org.apache.pig.tools.pigstats.ScriptState - Pig features used in the script: GROUP BY,ORDER BY,FILTER
2026-09-10 16:22:13,463 [main] INFO org.apache.pig.newplan.logical.optimizer.LogicalPlanOptimizer - {RULES_ENABLED=[AddForEach, ColumnMapKeyPrune, FilterConstantCalculator, ForEachConstantCalculator, GroupByConstParallelSetter, LimitOptimizer, LoadTypeCastInserter, MergeFilter, MergeForEach, NestedLimitOptimizer, PartitionFilterOptimizer, PredicatePushdownOptimizer, PushDownForEachFlatten, PushUpFilter, SplitConstantCalculator, SplitFilter, StreamTypeCastInserter]}
2026-09-10 16:22:13,513 [main] INFO org.apache.pig.impl.util.SpillableMemoryManager - Selected heap (PS Old Gen) of size 699400192 to monitor. collectionUsageThreshold = 489580128, usageThreshold = 489580128
2026-09-10 16:22:13,566 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MRCompiler - File concatenation threshold: 100 optimistic? false
2026-09-10 16:22:13,604 [main] INFO org.apache.pig.backend.hadoop.executionengine.util.CombinerOptimizerUtil - Choosing to move algebraic foreach to combiner
2026-09-10 16:22:13,622 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.SecondaryKeyOptimizerMR - Using Secondary Key Optimization for MapReduce
2026-09-10 16:22:13,642 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size before optimization: 3
2026-09-10 16:22:13,643 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.MultiQueryOptimizer - MR plan size after optimization: 3
2026-09-10 16:22:13,858 [main] INFO org.apache.pig.tools.pigstats.mapreduce.MRScriptState - Pig script settings are added to the job
2026-09-10 16:22:13,869 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.job.reduce.markreset.buffer.percent is deprecated. Instead, use mapreduce.reduce.markreset.buffer.percent
2026-09-10 16:22:13,869 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - mapred.job.reduce.markreset.buffer.percent is not set, set to default 0.3
2026-09-10 16:22:13,872 [main] INFO org.apache.hadoop.conf.Configuration.deprecation - mapred.output.compress is deprecated. Instead, use mapreduce.output.fileoutputformat.compress
2026-09-10 16:22:13,876 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Reduce phase detected, estimating # of required reducers.
2026-09-10 16:22:13,878 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Using reducer estimator: org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator
2026-09-10 16:22:13,885 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.InputSizeReducerEstimator - BytesPerReducer=1000000000 maxReducers=999 totalInputFileSize=345214189
2026-09-10 16:22:13,885 [main] INFO org.apache.pig.backend.hadoop.executionengine.mapReduceLayer.JobControlCompiler - Setting Parallelism to 1
```

Success!

Job Stats (time in seconds):

JobId	Maps	Reduces	MaxMapTime	MinMapTime	AvgMapTime	MedianMapTime	MaxReduceTime
job_local279989309_0001	11	1	n/a	n/a	n/a	n/a	n/a
job_local383886232_0002	1	1	n/a	n/a	n/a	n/a	n/a
job_local777184812_0003	1	1	n/a	n/a	n/a	n/a	n/a

SevereInjuryDistribution,

Input(s):

Successfully read 936812 records from: "C:/BDAL/BDAPProject/ChicagoTrafficCrashes.csv"

Output(s):

Successfully stored 5 records in: "C:/BDAL/BDAPProject/PIGFolder/SevereInjuryDistribution"

Counters:

Total records written : 5

Total bytes written : 0

Spillable Memory Manager spill count : 0

Total bags proactively spilled: 0

Total records proactively spilled: 0

Job DAG:

```
job_local279989309_0001 -> job_local383886232_0002,
job_local383886232_0002 -> job_local777184812_0003,
job_local777184812_0003
```

Output

```
C:\BDAL\BDAPProject\PIGFolder>dir SevereInjuryDistribution\*
Volume in drive C has no label.
Volume Serial Number is DA25-CDA0

Directory of C:\BDAL\BDAPProject\PIGFolder\SevereInjuryDistribution

09/10/2026  04:22 PM    <DIR>          .
09/10/2026  04:22 PM    <DIR>          ..
09/10/2026  04:22 PM                12 .part-r-00000.crc
09/10/2026  04:22 PM                8 ._SUCCESS.crc
09/10/2026  04:22 PM            129 part-r-00000
09/10/2026  04:22 PM                0 ._SUCCESS
               4 File(s)            149 bytes
               2 Dir(s) 129,432,944,640 bytes free

C:\BDAL\BDAPProject\PIGFolder>type SevereInjuryDistribution\part-r-00000
NO INDICATION OF INJURY|802596
NONINCAPACITATING INJURY|73983
REPORTED; NOT EVIDENT|41675
INCAPACITATING INJURY|15487
FATAL|1010
```

8. Conclusion

In this project, a large real-world Chicago traffic crash dataset was successfully prepared, stored, processed and analyzed using Hadoop HDFS, MapReduce and Apache Pig. Before analysis, the original dataset was reduced to 24 relevant columns and cleaned into a consistent CSV format so that it could be processed smoothly by Hadoop and Pig.

A total of 14 analyses were completed, including 6 MapReduce tasks and 8 Pig tasks. The MapReduce part focused on calculation-based analysis such as weather-wise crash counting, total injuries by road surface, average speed by crash type, injury rate, risk-score calculation and severe crash rate. Apache Pig was used for higher-level analysis such as grouping, filtering, sorting, ranking and trend analysis. The project examined traffic crashes from different perspectives, including weather, road surface condition, speed, injuries, traffic control, time, location, crash type and fatalities. These analyses helped reveal different patterns in the crash records and made the large dataset easier to understand. Some conditions and locations showed more crash or injury records than others, while the Pig analyses helped identify common crash causes, dangerous hours, monthly trends, frequent crash types and injury severity patterns.

Overall, the project demonstrated how big-data tools can be used to handle and analyze hundreds of thousands of records efficiently. Hadoop HDFS provided distributed storage, MapReduce handled large-scale calculations and Apache Pig made data analysis simpler through high-level operations. The project also provided practical experience in dataset preparation, HDFS file management, Java MapReduce programming, Pig scripting, command-line execution and interpreting large-scale traffic data.